

At The Border Of Viability: The Ethically Unique Case Of Extreme Preterm Infants

A patient's prognosis is a moving target, it juggles different variables in order to accurately predict when life will return back to a stable state. Exponential advances in modern medicine have contributed positively to this prognosis for a plethora of diseases and conditions. This has allowed humans to come back from odds that were once notoriously unfavourable; surgeons have faced death with a stable hand and scalpel. Extending the boundaries of viability has forced the law to adapt to a new instability; one charged with emotion, moral dilemma and ethical tension. A time pressure that was once isolated to the practice of medicine has been passed on to decisions of law, discussing the legal authority in end-of-life decisions whenever they may arise.

Facing the current limits of viability at the extremities of life, preterm infants are a triumph of modern medicine, and require particular consideration as to what defines an acceptable risk. Attempts at saving these tiny patients are being made as early as 22 weeks' gestation age, proving successful despite the ambiguity of the outcome or the potential for poor quality of life [1]. The conversations that lead to such decisions require more than empirical evidence and the hope for survival. They toy with parental authority, moral distress and legal considerations around what is in the best interest of the 'micro-preemies', as well as the physician's duty to a good practise of medicine. Hence, treatment of the innocent and premature has offered serious standards of controversy.

In my essay I bring to light why there is controversy regarding the intensive efforts of care preterm infants are subject to in their first few hours of life. I discuss the weight of parental autonomy, shared decision making, a practitioner's commitment to beneficence and non-maleficence, as well as the legal repercussions. Bringing U.S. case law as a means to ground

my arguments, I introduce *Miller vs HCA*, the subject of medical futility as seen with *In re Baby K*, and the Baby Doe regulations. Ultimately, my essay aims to encourage uncomfortable questions regarding the end-of-life, at birth, without undermining the difficult nature of these conversations.

The developmental process of human life in a mother's womb, from conception to birth, is measured in weeks, which are then grouped into three 13-week semesters [2]. Each trimester brings about irreplaceable progress in the creation of a healthy embryo, with organs and senses forming in a carefully conserved order as the fetus matures. By this timeline, a pregnancy to term is traditionally completed at around 39 weeks [2]. However, there are many instances that cause or require premature births, for instance, age, drug use, health conditions, abnormalities in the mother's cervix or uterus to name a few [3]. According to the World Health Organisation, a birth is officially considered premature when it occurs before the 37th week of pregnancy, and extreme prematurity is categorised as infants born before the 28th week of gestation [4]. There are an estimated 13.4 million neonates born preterm every year, born at varying extremities of maturity [4]. The 22-week mark is the earliest stage of fetal maturity, defined as periviable; it is the benchmark for where current medicine demonstrates a reasonable likelihood of extrauterine survival [2]. By the traditional timeline, at 22-weeks the limbs are developed and move in a coordinated manner, with the fingers demonstrating a strong grip [2]. The lungs have not fully developed, there is no melanin or body fat, and the fetus is incapable of survival without intensive care [5]. Yet, knowing the possibility of successful survival, it seems to contradict the Hippocratic vision of medicine to consider survival efforts subjective in the case of extreme premature infants.

At its core, medicine is about preserving a high quality of life. It does so with four cardinal principles that have adapted to the pace of modern medicine and moral priority: beneficence, non-maleficence, autonomy and justice [6]. Now a loose guideline, they remain foundational in situations of doubt and crucial to controversial circumstances, like those of premature infants. Therefore, it is often assumed that working toward the child's survival is in their best interest, a universal attempt at resuscitation with preserving life seen as a success of medicine. However, this model appears too binary, and is inconsiderate of quality of life, for, children who are born at 22-26 weeks gestational inequitably require life-saving treatment from physicians [5]. They are unable to survive outside the womb independently, which causes intensive resources and efforts to be performed [5]. Such extreme measures subject these infants to a future of adversities, usually involved in routine invasive procedures, severe morbidities, as well as physical and mental disabilities [7]. Hence, a conversation opens about quality of life, its costs, and if the most appropriate plan of action is one of aggressive intervention or peaceful passing.

Studies have demonstrated subjectivity in results when it comes to the outcomes of extreme preemies, contributing to their ethically unique nature. This challenges any standardised policy suggestions or universal standards of care, instead provoking a moral grey area. In July of 2025, a large meta-analysis across various databases was performed by Getaneh et al. in order to help relieve some instability [8]. They uncovered that income levels account for a drastic majority of this variability, providing survival of extreme preterm infants scale from over 90% to 10% in low-income countries [8]. Furthermore, disparities within countries were also identified as standards of healthcare, the definition of viability and attitudes towards lifesaving support were deemed nonuniform [9]. Lacking an even starting point for conversations regarding

attitudes toward extreme neonatal attitudes explains a portion of the discomfort in clinical decision-making. Not being grounded in statistics compromises attempts at successful prognosis, it makes shared decision making between parents and physicians more likely to be emotionally driven.

For extreme preemies who survive with the help of heroic intervention, they face high risks of brain damage and major short-term morbidities [10]. These include bronchopulmonary dysplasia, necrotizing enterocolitis, severe intraventricular haemorrhage, and severe retinopathy of prematurity [10]. An observational study, published by the Journal of American Medical Association, inquired into the 2 year neurodevelopmental outcomes of over 10,000 infants who were considered extreme preterm from 2013 to 2018 [11]. Just under half of those born before 27 weeks' had been rehospitalised, and 21.2% had severe neurodevelopmental impairment [11]. A long-term outlook raised concerns of cognitive impairment, cerebral palsy, blindness, deafness and compromised quality of life [10]. While clearly at risk of these conditions, there is no consistent predictor that can be applied during birth that would help refine prognosis.

Being sick claims a physical and economic toll; there is a price to pay for health. Those born in need of the Neonatal Intensive Care Unit (NICU) are no exception, with a 2021 analysis claiming that babies born at 22–28 weeks of gestation or critically ill require \$3,741 worth of care, daily [12]. An admission for babies who are considered critically ill or extremely premature can spike up beyond \$128,000 [12]. Furthermore, there are also instances where the pregnancy was initiated by expensive measures such as in vitro fertilisation (IVF), increasing the overall costs. The average couple who undergo IVF will not prove successful on their first attempt, accumulating an estimate of 2.3 - 2.7x IVF cycles in total [13]. Depending on location, company, or other variables, this could prove to be a \$50,000 investment, which often bestowed a “rescue”

attitude to further make physical, emotional, and financial burdens meaningful [13,14]. However, the fertilisation and NICU expenses are just the beginning. Children who are extremely premature require medical checkups as well as accommodations for any social, educational, or physical disability that often accompanies their survival [14]. These high costs can create distress and strains on families, and therefore play a factor when discussing withdrawal or withholding of life-sustaining therapy.

In situations where there is an imminent need for the birth to occur at the current border of viability, physicians need to discuss the burden of resuscitation as a feasible option, aligned with the parent's wishes and their clinical duty. This communication is often tainted with high emotions, a strict countdown, and a clear uncertainty. This was the case for Sidney Miller, born in Texas at 23 weeks gestational age who became a landmark to neonatal medical law as *Miller v. HCA* [15]. Anticipating an early delivery, Sidney's parents were informed of the unfortunate survival prognosis by their obstetrician and the hospital's neonatologist [16]. They explicitly addressed the near certainty that their child would develop profound disabilities, allowing Mark and Karla Miller to refuse resuscitative efforts on their infant daughter [16]. This decision was informed, clear, and documented: no neonatologist would need to be in the room as no "heroic intervention" would occur [16]. Despite this, the internal hospital administration overrode their decision, having a strict birth-weight dependent resuscitation policy intervene [16]. This meant that for Sidney who was born at 615g with a heartbeat, extreme measures would be performed without parental consent [16].

Although seemingly stable at birth, baby Sidney soon after suffered an intracranial hemorrhage which proved lifelong developmental damage [16]. This led her parents to sue the hospital, feeling the doctors only obliged to policy, and its larger company, HCA, for battery and

negligence from the absence of consent [15]. Seven years after her birth Sidney appeared in court where her condition was described as: “[S]he was 7 years old and could not walk, talk, feed herself, or sit up on her own. [She] was legally blind, suffered from mental retardation, cerebral palsy, seizures, and spastic quadriparesis in her limbs. She could not be toilet-trained and required a shunt in her brain to drain fluids that accumulate there and needed care 24 hours a day. The evidence further demonstrated that her circumstances will not change.” [16]. The Millers won their hearing and the judge concluded that both HCA and the hospital were negligent, awarding the Millers \$29,400,000 for medical expenses, \$17,503,066 in interest on these expenses, and \$13,500,000 in punitive damages [16]. Unfortunately for the Millers, the Texas Court of Appeals reversed the jury verdict ordering the family to receive no monetary compensation [16].

For the Texas Supreme Court, the legal debate at hand was not centered around viability or the permissibility of resuscitation, but rather on who held decisional authority during birth when prognosis had not been confirmed or discussed prior. The court initially confirmed the general principle: allocating the parents as the presumed decision maker for the actions of medical care of their children [16]. This includes the right to refuse treatment if considered medically reasonable and correctly informed [17]. However, it was determined that a consequential exception had occurred under emergent circumstances [16]. This attributed the authority back to physicians, regardless of parental consent, in order to initiate life-sustaining treatment to a newborn whilst avoiding any delays that could result in immediate death. It was ultimately reasoned that cases of extreme prematurity pose an ethically, medically, and legally unique aspect, with the inability to confidently determine what an “emergency” consisted of until the time was taken to clinically evaluate the infant once born [16]. Therefore the prognosis that

had been discussed between the Millers and the medical professionals was not deemed meaningful as such assessment was unachievable prior to birth [16]. This concluded their decisions as mere speculations rather than a legally binding mechanism of action.

The case of Sidney Miller and its ruling exposes the profound tension at the border of viability. Recognising the formal capacity of parental autonomy, it deconstructs a margin of exception during the narrow window of birth at which resuscitation is framed as an emergency intervention. The law subordinates this power to the doctor's discretion under a split-second decision, which justifies aggressive attempts at life preservation and survival. The default mandate to resuscitate demonstrated in *Miller v. HCA*, provides the risk of eradicating personalised medicine for the extreme premature cohort. It discredits the parents consent and decision-making capacity, ignoring the quality of life and long-term costs. The situation the Millers faced exemplifies how advances in neonatology have surpassed the legal and ethical stances on consent. This race of progression has left families with emotional and financial burdens, with children suffering through a life they are not able to survive in independently [18]. Conclusively, the discussions uprooted in *Miller v. HCA* have exemplified how the law is unprepared to confront extreme prematurity, prioritizing archaic foundations over the best intention of the most vulnerable patients.

Where *Miller v. HCA* uncovers how willing the law is to override parental refusal at the discretion of the physician during birth, *In re Baby K* reveals the opposite end of the scale. It exposes the tension when parents persist in the demand for extreme measures against the physician's authority. Escalating to federal court in 1994, the case of *Baby K* involves an infant born in October of 1992 with anencephaly rendering her permanently unconscious [19]. The congenital malformation meant that *Baby K* was missing a large part of her brain, skull and

scalp, causing her to have no cognitive abilities: unable to see, hear or interact with her environment [19]. Departing from debates on viability, Baby K presented a case where prognosis was medically certain rather than speculative [19]. Though she had the capability to breathe, with her autonomic functions supported by her brain stem, mechanical ventilation was required to stabilise her condition [19]. This removed any urgency, providing an opportunity for the parents and doctors to establish once more the diagnosis, prognosis, and plan of action, however, they failed to reach an agreement. Hence, although *Baby K* was not born at an extremely premature stage nor an uncertainty in prognosis, she forced the law to confront futility [19].

Physicians came to understand that treatment would have no curative conclusion or palliative purpose, recommending supportive care instead [19]. This meant nutrition, hydration, and a regulated heat source to keep *Baby K* comfortable [19]. Information regarding the situation was relayed to the parents and met with objection, introducing the fundamentals of medical futility [19]. These were used to justify the decision to not pursue further aggressive treatment regardless of the parents request. *Baby K*'s mother insisted on persisting with harsh measures to tackle any respiratory distress, leading to consistent admissions to the hospital who, on their end, sought out declaratory relief [19]. They argued it was both medically and morally inappropriate to administer treatment in the case of *Baby K*, and the physicians should not be legally compelled to oblige [19]. However, the court rejected the claim under the Emergency Medical Treatment and Active Labor Act, which positioned the hospital in a mandate to provide stabilizing treatment regardless of prognosis [19]. Furthermore, the hospital and its physicians were prohibited from denying treatment on the basis of disability under the Americans with Disabilities Act and the Rehabilitation Act [19]. Ultimately, acts like those listed above, or the 2001 Born-Alive Infants Protection Act, emphasise the requirement of intervention if a child is

born alive, regardless of their extent of development [9]. Having this mandate creates an increase in rigidity to instances of personalised medicine.

This provides a stark contrast to *Miller v. HCA*, where emergency doctrine was a means to empower physicians and surpass parental autonomy. *Baby K* shows how federal statutory law shapes treatment plans dismissing clinical input in favor of the conservative argument regarding end-of-life. Despite these differences, Sidney Miller and *Baby K* expose how the law favours the ease of decisions regarding end-of-life at the extremes of neonatal care. They uncover how there is limited application of medical futility to be enacted, weighed down by statutes prioritizing indiscriminate survival. As a result, the default outcome of legal decisions returns to the Hippocratic slogan of “do no harm”, where allowing for death to occur would present the most definite form of harm, irrespective of suffering, prognosis, or long-term quality of life. *Baby K* teaches lessons in discomfort for when the law faces an ethical gray zone. Her story, concluded by legal obligation, illustrates the imperative need for uncomfortable discussions around care for infants facing end-of-life decisions.

The enforced rigidity received by *Miller v. HCA* and *In re Baby K* is not coincidental, but rather the product of a broader regulatory framework introduced through Baby Doe regulations. Designed to prevent the withholding of treatment on the basis of disability, it was implemented after the passing of Baby Doe in 1982 [20]. Born with Down syndrome and tracheoesophageal fistula, Baby Doe’s death became highly publicised after the procedure to repair the tracheoesophageal fistula was actively opposed by the family [20]. This gathered outrage regarding the withholding of life-sustaining treatment on the basis of subjective judgment regarding quality of life [20]. Further enforced through amendments to the Child Abuse Prevention and Treatment Act, the regulations address any failure to provide medical treatment

on the basis of disability [20]. Defining such circumstances as a form of neglect, with very few exceptions, the Baby Doe regulations institutionalized a presumption of treatment [20]. This positioned survival as the legal baseline, inclusive of cases at the border of viability and regardless of medical futility.

The Baby Doe regulations provoked a lot of movement for both the legal and medical world: the definition of child abuse was tailored to include "failure to provide medically necessary treatment," and the standard that no prospect handicap should be considered in treatment decisions was reaffirmed with minimal exceptions [21]. Exceptions where it was considered legally permissible to withhold treatment of infants included: an irreversible coma, when treatment was shown to only prolong dying, or when treatment didn't prove to ameliorate the infant's condition [20]. Nevertheless, proven to be interpreted in a subjective manner, the cases of *Miller* and *Baby K* demonstrate a tendency to favor the conservative lens over clinical discretion or parental claim. The prognostic uncertainty that falls upon extreme prematurity, often prevents these scenarios to be classified as one of these exceptions, paving the way for statutory nondiscrimination acts to group these unique situations under inaccurate assumptions. This constrains cases of medical futility that include disability, which is a common occurrence when bordering on viability. Ignoring this distinction transforms ethical nuance into rigid regulations that encourage aggressive intervention and the burden that accompanies it.

When compared to *Miller v. HCA*, the Baby Doe regulations provide clarity on why universal resuscitation at birth is given priority in the law. Physicians have been taught to act in the means of conserving life whilst shielding allegations of neglect [22]. *Baby K* reveals how these same principles continue to persist beyond the immediate birth, an extension that extreme premature infants fall under, considering their potential for complications. Throughout these

cases and the indeterminacy of those born before 27 weeks gestational age, the Baby Doe regulations provide a foundation that leaves little room for case subjectivity. The appropriate action of shared decision making is no longer respected, and the power that already imbalances the relationship between a patient or surrogate, and the physician, is apparent if not emphasised. In a world where personalised medicine is valued, and the success of patient specificity is recognised, controversy arises for neonatal cases bordering on viability who are not awarded the same grace. When the law is renowned to be meticulous, why is the distinction between preserving life and prolonging dying...blurry? Having situations where, neither parental autonomy nor medical judgement, can justify the withdrawing or withholding of extreme medical efforts in order to allow a peaceful death, exposes a system that hasn't obtained the courage to address uncomfortable topics.

There is legitimate concern for advocacy regarding disabled infants, their vulnerability creates a potential for discriminatory decisional outcomes. The Baby Doe case proved, on the one hand, that medical futility could act as a front for withholding treatment of an infant diagnosed with Down syndrome. This case, alongside the sinister potential for eugenics, underscores the need for regulation and legislation advocating for presumption of treatment. Throughout this essay, the concept of "quality of life" proves a point regarding the subjective experience of a good life. In itself, this claim is harmful and inherently deems life with adversities, in whatever form they may be, as less worthy. This creates a slippery slope that medicine and factual science must avoid to prevent bias on life-changing decisions. This concern is found for preemies who have no personal autonomy, only loose predictions of neurodevelopmental outcomes. There is a certainty of the uncertainty of extreme preterm infants, suggesting that no universal framework can appropriately accommodate every situation as safely

as the default instinct of preserving life. The unavoidable ambiguity and fear of irreversible misstep justifies the argument persistently preserving life at the moment of birth. There is a looming ‘what if’ that can impose guilt of ‘giving up’.

The American legal system’s attitude towards decisions regarding extreme prematurity is notable when presented against international frameworks. In the case of Denmark, the country has embraced the ethically unique case of extreme preterm infants with a philosophy that has been in place for over two decades [23]. The Danish have bounced off of the unknown, allocating value to individual decisions and a personalised plan-of-actions [23]. Rather than mandating extensive resuscitation efforts at birth through immediate intubation, Danish practitioners use nasal continuous positive airway pressure which provides them with a stable window for a rational assessment of viability [23,24]. Known as the “wait-and-see”, the selective use of resuscitation creates a trial period to expose the capacity for survival before intrusive mechanical ventilation [23]. Where Denmark’s legal system notably diverges from the U.S. alternative, is in positioning the treating physician as the final decision makers, returning the ultimate power back to medicine [23]. Emphasising shared-decision making and the good practise of parent communication, however if the consent is not aligned with what the doctor believes to be in the best interest of the child, the professional can proceed with consent from the Child Welfare Board of the municipality [23]. The Danish attitude to actions on behalf of extremely premature infants has proved that the boundary between preserving life and prolonging dying can be sharper when the law returns clinical experience and knowledge to guide the margins of viability.

The common theme that is tied to viability is the understanding that it is a moving target, like the prognosis of a medical outcome. The border that medicine currently claims is notably

temporary, with inconsistencies in definition, statistical data, and according to location. Unable to provide stability through certainty, legal frameworks have been forced to neither commit to patient autonomy nor clinical judgment. Resisting the complexity of cases like *In re Baby K* or Sidney Miller, the law has retreated to rigid mandates under a broad desire of survival, regardless of nuance or quality. By framing resuscitation as a universal requirement, the resulting system is one that shifts further burdens onto families and infants who are already facing adversities.

The conclusion to my essay is not one against universal resuscitation at the border of viability, nor absolute parental control. It is a means to open up a conversation about the ethically unique case of extreme prematurity. A situation that, in the U.S., currently faces legislation that defaults to the safety of survival, inconsiderate of any ifs, buts, or maybes. Collapsing at uncertainty exposes how necessary these discussions are. The law must move with medical advances instead of proving paralysed at the progress. The children born at 22, 23, 24 weeks deserve more than institutional self-protection, they deserve decisions made with careful understanding, compassion, and an honest confrontation as to what their lives, and their suffering, will bring about.

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